Fast Inverse-Scattering Reconstruction for Airborne High-Squint Radar Imagery Based on Doppler Centroid Compensation

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***Abstract*— Cross-resolution enhancement for airborne high-squint radar (AHSR) imagery is mathematically equivalent to the ill-conditioned problem of inverse-scattering reconstruction. Although a variety of inversion methods with regularization can be introduced to advance the field of AHSR imagery, they turn out to be computationally intensive when extended to 2-D (range and cross-range dimension) image formulation due to the range-by-range calculation for the space-variant inversion operators over the full range swath. To tackle the problem of efficiency, this article presents a low-complexity inverse-scattering strategy. Our underlying idea is to equalize the space-variant Doppler centroid embedded in an inversion operator for a reference range cell using Doppler centroid compensation. With the proposed strategy, the necessary computational complexity required for 2-D AHSR inverse-scattering reconstruction can be significantly reduced by requiring only the calculation of the inversion operator, independently of the number of range cells. Our experimental assessment, conducted using both the simulation and real data, demonstrates that our proposed inverse-scattering strategy offers preferable computational reduction in the task of inverse-scattering reconstruction for 2-D AHSR imagery without resolution loss.**

***Index Terms*— Airborne high-squint radar (AHSR), cross-resolution, Doppler centroid compensation, inverse-scattering, space-variant.**

I. Introduction

AIRBORNE high-squint radar (AHSR) imaging has raised many concerns in the fields of both civilian and military applications, such as terrain awareness and aircraft self-landing. When using AHSR imagery for these applications, sufficient 2-D resolution is required to recover fine descriptions of the observed scenario. Often, the desirable slant-range resolution can be obtained by transmitting wideband signals, whereas the cross-range resolution is usually improved by the Doppler beam sharpening (DBS) method [1], [3], [4]. The principle of DBS is to distinguish the linear components of Doppler history generated by the relative movement of the radar platform and targets with different azimuths in the high-squint direction. To achieve this, narrow-band filters implemented using the classical fast Fourier transform (FFT) are widely adopted. Often, one has to resort to a large coherent processing interval (CPI) to achieve a desirable cross-range resolution [5], [6]. Regrettably, there are various limitations that may prevent choosing a large CPI, such as high-order Doppler effect, range walk, and target scintillation [7]–[9]. This motivates the development of methods that can improve cross-range resolution of AHSR imagery when the CPI is limited.

Because the fundamental task of DBS is to resolve the Doppler spectrum embedded in the echo, it makes sense to use advanced spectral-estimation methods to enhance the cross-range resolution. Wu *et al.* [5] incorporated aperture extrapolation (AE) with the amplitude and phase estimation (APES) algorithm to improve the cross-range resolution of AHSR images. It was shown that a higher cross-range resolution than that possible for the FFT-based DBS images can be achieved. Such an implementation, however, requires many snapshots, as well as uncorrelated scatterers, conditions that are rarely satisfied in practical radar imaging [10]. To overcome this problem, the iterative adaptive approach (IAA) [11], [12] was introduced [13], [14], and it was shown that the IAA can work with few or even a single snapshot as well as correlated scatterers, and can provide higher resolution and lower sidelobe levels with respect to the FFT method.

More recent developments in radar imaging theory suggest that the cross-resolution enhancement for AHSR imagery is mathematically equivalent to an ill-conditioned problem of inverse-scattering reconstruction, whereby a spatial map of reflectivity may be reconstructed from noisy measurements with substantial improvements in the resolution and quality of the processed images [10]. Like many other situations for microwave sensing, the problem of the inverse-scattering problem for AHSR imagery is ill-posed, and direct inversion

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algorithms, such as the classical least squares (LS) method, will be highly noise-sensitive. Based on the established inverse-scattering model, a variety of inversion methods with regularization can be introduced to solve the ill-posed problem. At least two classes of algorithms with a regularization procedure appear in existing microwave sensing applications. The first kind is based on the theory of singular value decomposition (SVD), which expresses the LS reconstruction result as a linear combination of the singular values. It is demonstrated that the smaller singular values are strongly affected by the noise. With this insight, the truncated SVD (TSVD) algorithm was developed to solve the ill-posed problem; it obtains a stable solution by properly excluding the smaller singular values [15]–[19]. Another type of regularization method is to limit the range of its solution space by imposing various priors as regularization terms on the considered spatial map of reflectivity. The most basic method is based on the l_2-norm, which is the famous Tikhonov regularization [20]. The procedure has been shown to achieve a spatial resolution improvement, but still suffers from a high level of side lobes. To obtain further resolution improvement, researchers have surveyed the roles of sparsity in existing radar imaging practice [7], [10]. Such sparsity offers a means for regularizing the inverse-scattering task by imposing an l_0-norm constraint on the spatial map of reflectivity. Successful applications in the field of radar imaging can be found in [21]–[23]. Because solving the l_0-norm penalized problem is typically infeasible in combination, many classical sparse methods, such as the least absolute shrinkage and selection operator (LASSO) [24] and the focal underdetermined system solution (FOCUSS) [25], have focused on formulating convex algorithms that instead exploit l_1-norm penalties. The potential drawback of the LASSO and FOCUSS is the selection of one or more hyperparameters, which would often be a nontrivial task. To relax the requirement of user parameters, a novel sparse method, termed the sparse iterative covariance-based estimation (SPICE) algorithm, was proposed based on covariance fitting criteria [26]–[29]. The SPICE algorithm is globally convergent and does not require any choice of user parameter. Other extensions of the regularization methods are incorporated with edge characteristics of the considered spatial map of reflectivity, and the regularization terms are commonly combined with the total variation (TV) function to improve the reconstruction quality of the edge [30].

Computational complexity becomes a critical issue when applying inversion algorithms to high-dimensional data. Currently, there are three main kinds of developments in fast implementation to mitigate the computational burden of brute-force inversion methods. The first class is based on the rich matrix structure of the problem [31]–[37]. For example, when the involved matrices have a Toeplitz or Vandenmonde structure, the computational complexity of matrix operations may be notably reduced by using the Gohberg–Semencul representation and the FFT [31]. The second class involves the adoption of a more efficient solver. The well-known conjugate gradient algorithm is a representative fast solver for the LS problem that avoids the forming of a direct matrix inverse. Other examples, including the split-Bregman and iteratively reweighted linear least squares (IRLS) methods, can efficiently solve the class of l_1-regularized optimization problems [38]–[40]. The third class involves exploitation of the correlation between adjacent data samples to formulate a recursively updated solver, which is more suitable for online processing, such as the classical recursive LS (RLS) algorithm and online methods for other typical applications [41]–[47].

However, even considering these techniques, further improvements on the current advances on fast inversion methods are still relevant in the context of 1-D processing. In practice, an AHSR image needs to be formulated in both the range and cross-range direction, whereby the sampled echo matrix is first processed by pulse compression in the range dimension, with the inversion methods then being applied to enhance the cross-range resolution for each range cell. As will be shown in this article, the computationally extensive inversion operators are defined by the space-variant Doppler centroid, requiring them to be calculated for each range cell. This means that the computational complexity quickly becomes cumbersome when considering a large number of range cells.

In this work, we develop a novel, low-complexity, inverse-scattering approach for 2-D AHSR images. The key idea is to equalize the space-variant Doppler centroid embedded in the inversion operator for a reference range cell. This allows the space-variant inversion operator to be formulated so that one is only required to form a single inversion operator in oder to realize the inverse-scattering reconstruction over the full range swath. We call this process Doppler centroid compensation. The resulting strategy allows for a significantly reduced of the computational complexity, independent of the number of considered range cells.

The remainder of this article is organized as follows. In Section II, we review the inverse-scattering model for AHSR imaging. In Section III, several alternative inversion methods are introduced for recovering the considered AHSR imagery. In Section IV, the proposed low-complexity inversion strategy for 2-D AHSR image formulation is detailed. In Section V, some practical algorithmic considerations are discussed. In Sections VI and VII, simulation and real-data results are presented to demonstrate the resolution and computational merits in practical remote sensing applications. Finally, the conclusions are presented in Section VIII.

II. Background

The geometric relationship between the radar platform and the AHSR-imaged area in high-squint directions is illustrated in Fig. 1. The reference Cartesian coordinate is defined by $Oxyz$. Suppose that the platform flies at velocity v along the x-axis. The instantaneous slant distance between the target and the platform is defined as $R(t)$, where t is the time variable in the cross-range dimension (slow time), and R_0 is the initial slant range for $t = 0$. The azimuth and pitch angles are defined by θ and φ, respectively, whereas H denotes the flight height of the platform.

Suppose that the radar transmits a linear frequency-modulated (LFM) signal

$$f(\tau) = \text{rect}\left[\frac{\tau}{T_p}\right] \exp\left(j2\pi f_c\tau + j\pi\gamma\tau^2\right) \tag{1}$$

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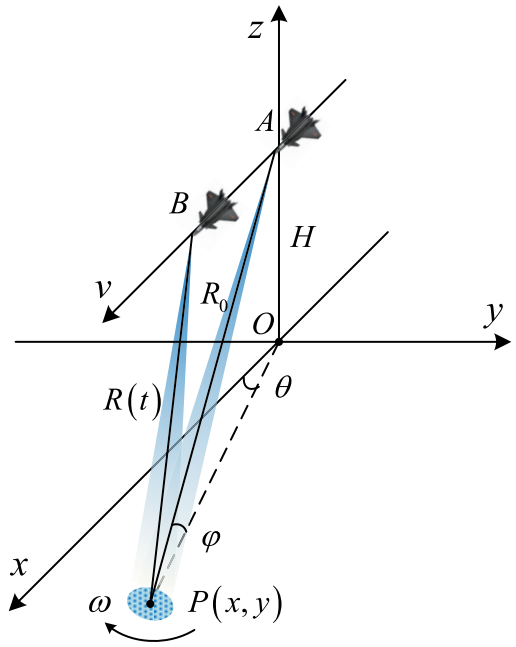


Fig. 1. AHSR imaging geometry.

where τ denotes the range time variable, T_p the pulsewidth, f_c the carrier frequency, exp(·) the exponential function, and γ the chirp rate, an where the function rect(·) is the rectangular pulse envelope, defined as

$$\mathrm{rect}\left(\frac{\tau}{T_p}\right)=\begin{cases}1, & |\tau|\le\dfrac{T_p}{2}\\ 0, & \text{otherwise.}\end{cases} \tag{2}$$

After demodulation, range matched filtering and range migration correction [48], the continuous noise-free signal model of the received echo for a point target can be expressed as

$$y(\tau,t)=\kappa\cdot\sigma\cdot h\left(t-\frac{\theta}{\omega}\right)\mathrm{sinc}\left[B\left(\tau-\frac{2R_0}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi}{\lambda}R(t)\right] \tag{3}$$

where κ denotes a constant that reflects various gains and losses caused by, for instance, wave spreading, atmospheric attenuation losses, which causes radiation changes in the range dimension of the image, σ is the effective scattering cross section of the target located at the point A, and $h(t)$ the two-way antenna pattern function. It is worth noting that the formulation neglects the effect of the antenna elevation pattern, because we primarily consider the processing of echoes for each range cell, where the antenna elevation pattern would be constant along the cross-range direction. This simplification will not affect the processing in the cross-range direction. Furthermore, in (3), ω denotes the scanning speed, c the velocity of the electromagnetic wave, λ the transmitted signal wavelength, and B the bandwidth of the transmitting signal. The function sinc(·) is defined as $\mathrm{sinc}(x)=\sin(\pi x)/\pi x$.

High range resolution can be obtained by pulse compression, which is defined by the Sinc function term in (3). However, the cross-range resolution is related to the actual beamwidth of the antenna pattern, $h(t)$. By exploiting the Doppler shift generated by the relative movement of the radar platform and targets with different azimuths, a higher cross-range resolution can be obtained via the DBS method.

A. DBS Theory and Methods

According to the geometry shown in Fig. 1, the instantaneous slant range for a point target is given by

$$R(t)=\sqrt{R_0^2+(vt)^2-2R_0vt\cos\theta\cos\varphi}. \tag{4}$$

Expanding $R(t)$ as a Taylor series, one obtains

$$R(t)=R_0-vt\cos\theta\cos\varphi+\frac{v^2\cos^2\theta\cos^2\varphi}{2R_0}t^2+O\left(t^2\right) \tag{5}$$

where $O(t^2)$ represents the higher-order term of t^2. For the case of $vt\ll R_0$, the quadratic and higher-order terms in (5) can be ignored [22]. Thus, (5) may be well approximated as

$$R(t)\approx R_0-vt\cos\theta\cos\varphi. \tag{6}$$

Substituting (6) into the Doppler term in (3) yields

$$\exp\left[-j\frac{4\pi}{\lambda}R(t)\right]=\exp\left[-j\frac{4\pi}{\lambda}R_0\right]\cdot\exp[j2\pi f_dt] \tag{7}$$

where

$$f_d(\theta,\varphi)=\frac{2v\cos\theta\cos\varphi}{\lambda} \tag{8}$$

is the Doppler frequency for the azimuth angle θ and the pitch angle φ. If X narrowband filters are placed in the frequency-domain, with center frequencies aimed at the centroid of the Doppler spectrum in the cross-range direction, targets located in different cross-range directions may be distinguished.

In practice, X narrowband filters can be implemented using a X-point FFT operation. Correspondingly, a formula of a sharpening ratio can be derived, expressed by

$$\beta=\frac{\Delta f_d}{\delta_{\mathrm{FFT}}} \tag{9}$$

where

$$\Delta f_d\approx\frac{2v\sin\theta\cos\varphi}{\lambda}\Delta\theta \tag{10}$$

is the Doppler bandwidth within the mainlobe beamwidth [4], $\Delta\theta$ the mainlobe width of the antenna pattern, and δ_{FFT} the frequency resolution of the FFT. Following the derivation in [21], the sharpening ratio may be expressed as:

$$\beta=\frac{2\pi v\Delta\theta f_c\sin\theta\cos\varphi}{180c}\cdot\mathrm{CPI} \tag{11}$$

where $\mathrm{CPI}=\Delta\theta/\omega$.

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B. Problem Formulation

As seen from (11), increasing the CPI is an effective way to improve the cross-range resolution. However, when the CPI is increased, the condition $vt \ll R_0$ may not be satisfied, and the higher-order terms in (5) may not be ignored. Meanwhile, to guarantee the wide-area surveillance and high-frame real-time imaging, there are limitations to how long a DBS mode can collect pulses while maintaining coherence across a single acquisition. Other limitations, such as range walk and target scintillation will also prevent choosing a large CPI [8], [9].

Instead of increasing the CPI, a variety of advanced approaches designed for spectral estimation can be applied to obtain higher cross-range resolution than the FFT method. However, many factors limit the quality of spectral estimates, such as a low number of snapshots, high levels of correlated clutter, and high computational complexity.

Model mismatch is another key problem that has not been well handled in modern systems. The current spectral-analysis model primarily considers linear Doppler frequencies, neglecting the effect of the antenna scanning shown in (3). Such approximations will also degrade the quality of the AHSR image.

III. AHSR Imaging Based on Inverse-Scattering Theory

To address the problems of current DBS imaging strategies, we first establish an inverse-scattering model, which accounts for the more accurate Doppler history and the effects of antenna scanning. Then, building on the recent developments in radar inverse-scattering theory, we combine this model with several inverse-scattering methods to improve the cross-range resolution of the AHSR imagery.

A. Inverse-Scattering Model

In range, the fast-time sampling is performed by analog-to-digital inversion, whereas slow-time sampling is performed by periodically transmitting pulses and receiving the echo at the rate of pulse repetition frequency (PRF). Based on this, the continuous 2-D signal expressed by (3) can be discretized into an $M \times N$ echo matrix

$$\mathbf{Y} = \begin{bmatrix} y(1,1) & y(1,2) & \cdots & y(1,N) \\ y(2,1) & y(2,2) & \cdots & y(2,N) \\ \vdots & \vdots & \ddots & \vdots \\ y(M,1) & y(M,2) & \cdots & y(M,N) \end{bmatrix} \triangleq \begin{bmatrix} \mathbf{y}_1^T \\ \mathbf{y}_2^T \\ \vdots \\ \mathbf{y}_M^T \end{bmatrix} \tag{12}$$

where M and N represent the number of sampling points in the range and cross-range direction, respectively, $y(m,n)$ the sampled value of the complex envelope, and the $N \times 1$ vector $\mathbf{y}_m = [y(m,1), y(m,2), \ldots, y(m,N)]^T$ denotes the cross-range slice at the mth range cell, for $m = 1, \ldots, M$.

Let Φ denote the scanning range. The number of sampling points in the cross-range dimension may then be expressed as $K = \mathrm{PRF} \cdot \Phi/\omega$. Suppose that each point of a predefined grid that covers the scanning range of interest is considered to be a potential target whose scattering coefficient is to be estimated. The cross-range slice, $\mathbf{y}_m$, for $m = 1, \ldots, M$, can then be modeled as

$$\mathbf{y}_m = \mathbf{A}_m \mathbf{s}_m + \mathbf{e}_m \tag{13}$$

where $\mathbf{A}_m$ represents the forward matrix, and $\mathbf{s}_m = [s_m(\theta_1), s_m(\theta_2), \ldots, s_m(\theta_K)]^T$ the vector of the reflectivity function at the mth range cell, with $s_m(\theta_k)$ being proportional to the radar cross section (RCS) of the scatter located at θ_k. The vector $\mathbf{e}_m \in \mathbb{C}^{N\times 1}$ denotes an additive white Gaussian noise (AWGN). By considering the discrete form of (3), the forward matrix is given by $\mathbf{A}_m = \mathbf{H} \odot \mathbf{D}_m$, with $\odot$ denoting the Schur–Hadamard product, and

$$\begin{aligned} \mathbf{H} &= [\mathbf{h}(\theta_1), \mathbf{h}(\theta_2), \ldots, \mathbf{h}(\theta_K)] \\ \mathbf{D}_m &= [\mathbf{d}_m(\theta_1), \mathbf{d}_m(\theta_2), \ldots, \mathbf{d}_m(\theta_K)] \end{aligned} \tag{14}$$

where

$$\begin{aligned} \mathbf{h}(\theta_k) &= \left\{ h\left[t_1 - \frac{\theta_k}{\omega}\right], \ldots, h\left[t_N - \frac{\theta_k}{\omega}\right] \right\}^T \\ \mathbf{d}_m(\theta_k) &= \exp\left\{ -j\frac{4\pi}{\lambda}[R_m(t_1), \ldots, R_m(t_N)] \right\}^T \end{aligned} \tag{15}$$

denotes the antenna and space-variant Doppler phase vectors of size $N \times 1$ for the target located at the azimuth angle θ_k, for $k = 1, \ldots, K$, respectively, R_m the slant range for the target located at the mth range cell. Here, the variables t_n, for $t_n, n = 1, \ldots, N$, denote the slow time samples. By fully considering the effects of the antenna scanning, the matrix $\mathbf{H}$ can be reconstructed with L samples of the antenna pattern, $h_1, h_2, \ldots, h_L$, such that

$$\mathbf{H} = \begin{bmatrix} h_L & h_{L-1} & \cdots & h_1 & & \\ & h_L & h_{L-1} & \cdots & h_1 & \\ & & \ddots & \ddots & & \ddots \\ & & & h_L & h_{L-1} & \cdots h_1 \end{bmatrix} \tag{16}$$

and

$$N = K - L + 1. \tag{17}$$

In the model shown in (13), both the effects of the antenna scanning and the Doppler history are fully taken into consideration, which characterizes the physical process of echo acquisition more precisely. This allows for more accurate reconstruction of the AHSR imagery when applying inverse-scattering methods to reconstruct the surface map of reflectivity.

B. Inverse-Scattering Reconstruction for AHSR Imagery

As shown in (13), the problem of AHSR imagery reconstruction can be mathematically expressed by a set of linear equations, allowing various inverse-scattering methods to be applied to reconstruct the AHSR image. The most basic way to solve (13) is to use the LS method, yielding

$$\hat{\mathbf{s}}_m = \left(\mathbf{A}_m^H \mathbf{A}_m\right)^{-1} \mathbf{A}_m \mathbf{y}_m. \tag{18}$$

for $m = 1, \ldots, M$. However, this solution is typically ill-conditioned, and the additive noise may result in severe perturbation of the estimate.

To overcome the problem, a variety of inverse-scattering methods with a regularization procedure have become increasingly important in several alternative microwave sensing applications. Successful recently applications can be found in,

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e.g., [15]–[23], [26]–[30] and the references therein. In addition to achieving noise suppression, the developed inverse-scattering methods can offer substantial resolution improvement. In view of these potentials, three classical algorithms are here introduced to deal with the problem of AHSR image reconstruction, which can be used to effectively enhance the cross-range resolution.

1) TSVD Method: Consider the SVD of the steering matrix $\mathbf{A}_m$, expressed by [30]

$$\mathbf{A}_m = \mathbf{U}\Sigma\mathbf{V}^T = \sum_{n=1}^{N} \sigma_n \mathbf{u}_n {\mathbf{v}_n}^T \tag{19}$$

where $\mathbf{U}$ and $\mathbf{V}$ are the matrices containing the singularvectors, $\mathbf{u}_n$ and $\mathbf{v}_n$, respectively, whereas $\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_N)$ denotes a diagonal matrix, whose diagonal elements are the singular values of $\mathbf{A}_m$. Using (19), the inverse-scattering results may be expressed as

$$\hat{\mathbf{s}}_m = \sum_{n=1}^{N} \frac{\mathbf{u}_n^T \mathbf{y}_m}{\sigma_i} \mathbf{v}_n^T. \tag{20}$$

It can be shown that the smaller singular values of σ_i are less reliable and it is often preferable to discard these [15]–[19], [30]. In the TSVD method, this is done via the choice of a rational parameter removing the small singular values, creating the low-rank approximation

$$\hat{\mathbf{s}}_m = \sum_{n=1}^{Q} \frac{\mathbf{u}_n^T \mathbf{y}_m}{\sigma_i} \mathbf{v}_n^T \tag{21}$$

where Q is the number of nontruncated singular values.

2) l_2-Based Method: The Tikhonov, or l_2-norm-based, regularization is formed by incorporating an *a priori* assumption about the exact solution in the seminorm $\|\Gamma\mathbf{s}_m\|_2$, where Γ is a regularization matrix [49]. This leads to the minimization problem

$$J = \|\mathbf{y}_m - \mathbf{A}_m\mathbf{s}_m\|_2^2 + \lambda\|\Gamma\mathbf{s}_m\|_2^2 \tag{22}$$

where the regularization parameter $\lambda > 0$ controls the tradeoff between the residual norm $\|\mathbf{y}_m - \mathbf{A}_m\mathbf{s}_m\|_2^2$ and the regularity of the solution $\|\Gamma\mathbf{s}_m\|_2^2$. When the regularization matrix Γ is an identity matrix, the solution to the problem (22) can be expressed as a diagonal loading, that is,

$$\hat{\mathbf{s}}_m = \left(\mathbf{A}_m^H\mathbf{A}_m + \lambda\mathbf{I}\right)^{-1}\mathbf{A}_m^H\mathbf{y}_m. \tag{23}$$

3) l_1-Based Method: Sparsity is an important prior for a considerable amount of cases [24]–[30]. Recent developments in radar inverse-scattering imaging suggest that sparsity can lead to further performance benefits by imposing sparsity as a statistical prior on the considered scenario. The basic sparse approach replaces the l_2-norm regularization term in (22) with an l_1-norm penalization instead, resulting in the classical LASSO estimator, which solves the following convex problem:

$$\hat{\mathbf{s}}_m = \arg\min_{\mathbf{s}_m}\left\{\|\mathbf{y}_m - \mathbf{A}_m\mathbf{s}_m\|_2^2 + \lambda\|\mathbf{s}_m\|_1\right\} \tag{24}$$

for $m = 1, \ldots, M$. The split-Bregman method is frequently employed to solve the optimization problem, allowing constrained optimization problems to be reduced to a short sequence of unconstrained LS problems using the Bregman iteration [38]. Compared to conventional Newton-type methods, the split-Bregman method has much faster convergence and a higher numerical accuracy [38]. The key idea of the split-Bregman algorithm is to separate different variables and cross-iteratively solve them by transforming (24) into an unconstrained optimization problem

$$\hat{\mathbf{s}}_m = \arg\min_{\mathbf{s}_m} \|\mathbf{y}_m - \mathbf{A}_m\mathbf{s}_m\|_2^2 + \frac{\mu}{2}\|\mathbf{d} - \lambda\mathbf{s}_m\|_2^2 + \|\mathbf{d}\|_1 \tag{25}$$

where $\mathbf{d}$ is an introduced temporary vector, and μ an additional regularization parameter. The improvement brought by the split-Bregman method is at the cost of adding one more user parameter, which leads to difficulties in parameter selection in practical problems. Fortunately, the recently proposed SPICE method converts (24) into a weighted square root LASSO problem, such that

$$\hat{\mathbf{s}}_m = \arg\min_{\mathbf{s}_m}\left\{\|\mathbf{y}_m - \mathbf{A}_m\mathbf{s}_m\|_2 + \|\Lambda\mathbf{s}_m\|_1\right\} \tag{26}$$

where Λ can be expressed as [28], [57], [58]

$$\Lambda = \mathrm{diag}\left(\sqrt{\frac{\|\mathbf{a}_1\|_2^2}{N}}, \sqrt{\frac{\|\mathbf{a}_2\|_2^2}{N}}, \cdots, \sqrt{\frac{\|\mathbf{a}_K\|_2^2}{N}}\right) \tag{27}$$

with $\mathbf{a}_k$ denoting the kth column of the matrix $\mathbf{A}_m$. With this development, the split-Bregman method can be used for sparse reconstruction with a single user parameter, such that

$$\begin{aligned}
\hat{\mathbf{s}}_m^{q+1} &= \left(2\mathbf{A}_m^H\mathbf{A}_m + \mu\Delta\right)^{-1}\left(2\mathbf{A}_m^H\mathbf{y}_m + \mu\Lambda^H\left(\mathbf{d}^q - \mathbf{b}^q\right)\right) \\
\mathbf{d}^{q+1} &= \mathrm{shrink}\left(\hat{\mathbf{s}}_m^{q+1} + \mathbf{b}^q, 1/\mu\right) \\
\mathbf{b}^{q+1} &= \mathbf{b}^q + \hat{\mathbf{s}}_m^{q+1} - \mathbf{d}^{q+1}
\end{aligned} \tag{28}$$

where q is the iteration number, $\Delta = \Lambda^H\Lambda$, $\mathrm{shrink}(x, \eta) = \mathrm{sign}(x)\max(|x| - \eta, 0)$, and

$$\mathrm{sign}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases} \tag{29}$$

denotes the sign function.

C. Computational Complexity for 2-D AHSR Image Formulation

The flowchart of the 2-D AHSR image formulation is shown in Fig. 2. Here, $\mathbf{A}_m$ is defined by the space-variant Doppler phase, $\mathbf{D}_m$. As a result, the required computationally extensive matrix operations, such as the SVD or matrix inversion in the representative inverse-scattering methods, need to be calculated for each range cell (see the red frame in Fig. 2). The resulting computational complexity is thus $\mathcal{O}(MN^3)$. Such notably high computational complexity renders the current methods intractable.

IV. Fast 2-D Inverse-Scattering Reconstruction Based on Doppler Centroid Compensation

The key idea of the proposed strategy is to remove the space-variant Doppler centroid embedded in the inversion operator, $\mathbf{A}_m$. In this way, the inversion operators for all

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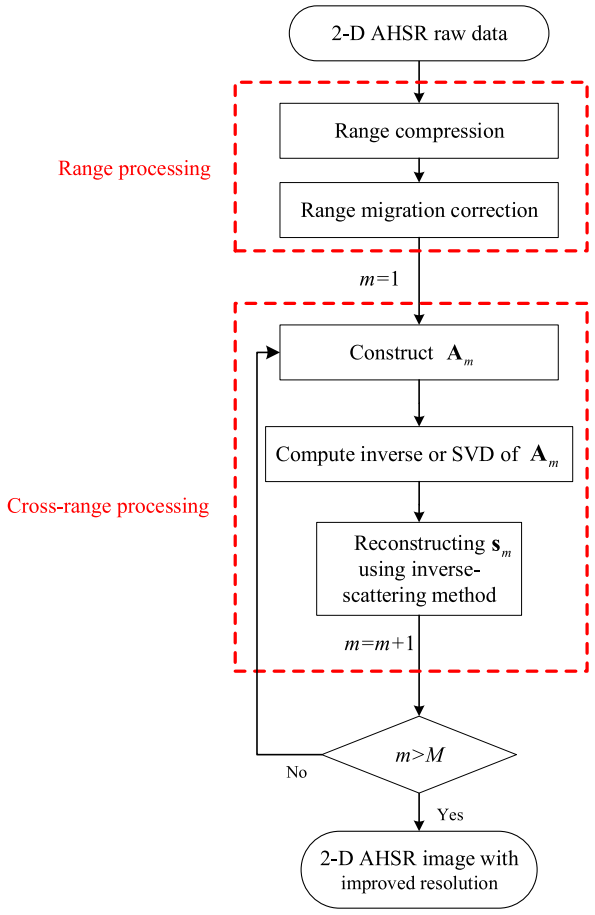


Fig. 2. Flowchart of 2-D AHSR image using inverse-scattering methods.

range cells can therefore be uniformly corrected, and only one calculation of inversion operators will be needed to formulate the 2-D AHSR image.

A. Basic Principle

As seen from (6) and (15), it is difficult to directly correct the space-variant term R_m in $\mathbf{A}_m$ to a constant value, as the term is embedded in the exponential power. Fortunately, the insight into the characteristics of the echo in the range-Doppler (RD) domain provides a feasible way to eliminate the space variance.

As shown in the left part of Fig. 3, when there is no space variation, the Doppler frequency for each azimuth angle would not change along the range direction. In this case, the shaded area within the Doppler bandwidth should be in the shape of a rectangle. In contrast, when the space variation exists, the Doppler frequency distribution within the Doppler bandwidth is modulated by a cosine function, as shown by the right part of Fig. 3. Therefore, if a method can be found to correct the space-variant spectrum to a rectangle, then the space variance can be reasonably removed.

B. Doppler Centroid Compensation

Suppose that $\mathbf{Y}_{\mathrm{RD}}$ is the matrix of the RD image obtained by performing a N_{FFT}-point FFT on the echo matrix $\mathbf{Y}$ in the

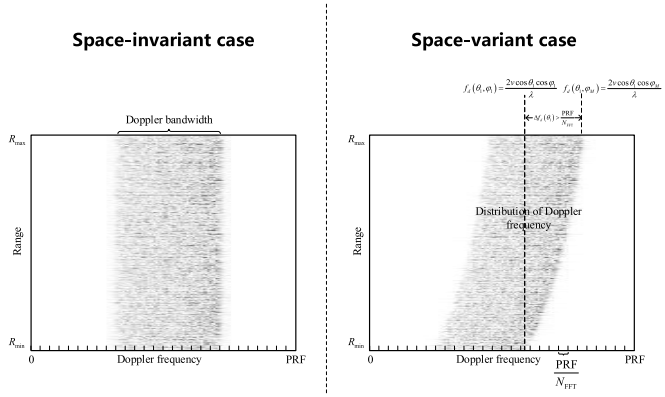


Fig. 3. Distribution of Doppler frequency in the RD domain for both the space-invariant and the space-variant case.

cross-range direction, where N_{FFT} denotes the smallest power of two that is closest to the absolute value of N. Before we proceed to correct the space-variant Doppler frequencies, it is necessary to determine the space variance in order to avoid unnecessary Doppler centroid compensation when the space-variant problem is not serious, thereby reducing the required computational cost.

1) Criterion for Space Variance: According to the FFT principle, the Doppler frequency cell may be defined as $\mathrm{PRF}/N_{\mathrm{FFT}}$. The maximum frequency difference $\Delta f_d(\theta_k)$ may be determined by the difference between the maximum and minimum range cell as

$$\begin{aligned}\Delta f_d(\theta_k) &= |f_d(\theta_k, \varphi_1) - f_d(\theta_k, \varphi_M)| \\ &= \left|\frac{2v\cos\theta_k}{\lambda}(\cos\varphi_1 - \cos\varphi_M)\right|. \end{aligned} \tag{30}$$

As shown in Fig. 3, the space variance exists when the frequency difference $\Delta f_d(\theta_k)$ exceeds more than one frequency cell $\mathrm{PRF}/N_{\mathrm{FFT}}$, such that

$$\Delta f_d(\theta_k) > \frac{\mathrm{PRF}}{N_{\mathrm{FFT}}}. \tag{31}$$

In this case, it is necessary to perform Doppler centroid compensation. On the contrary, if the $\Delta f_d(\theta_k)$ of all azimuths is less than $\mathrm{PRF}/N_{\mathrm{FFT}}$, the distribution of $\mathbf{Y}_{\mathrm{RD}}$ is originally within a rectangular area. This means that there is no obvious space variance, and the Doppler centroid compensation is not required.

2) Compensation: To correct the space-variant RD spectrum, we first need to select a group of reference frequencies, $f_d(\theta_k, \varphi_{\mathrm{ref}}), k = 1, \ldots, K$, as a relocation benchmark. It is recommended that the reference frequencies are selected as

$$f_d(\theta_k, \varphi_{\mathrm{ref}}) = \frac{f_d(\theta_k, \varphi_1) + f_d(\theta_k, \varphi_m)}{2}, \quad k = 1, \ldots, K \tag{32}$$

As shown in Fig. 4, the specific operation of compensation is actually to correct the curvature of the Doppler distribution $f_d(\theta_k, \boldsymbol{\varphi})$ (solid line) to $f_d(\theta_k, \varphi_{\mathrm{ref}})$ (dashed line). The lines of different colors indicate different azimuth angles. Since these lines are composed of data points in the Doppler frequency domain, we need point-by-point projection. In term of the reference frequency, $f_d(\theta_k, \varphi_{\mathrm{ref}})$, the offset between the reference

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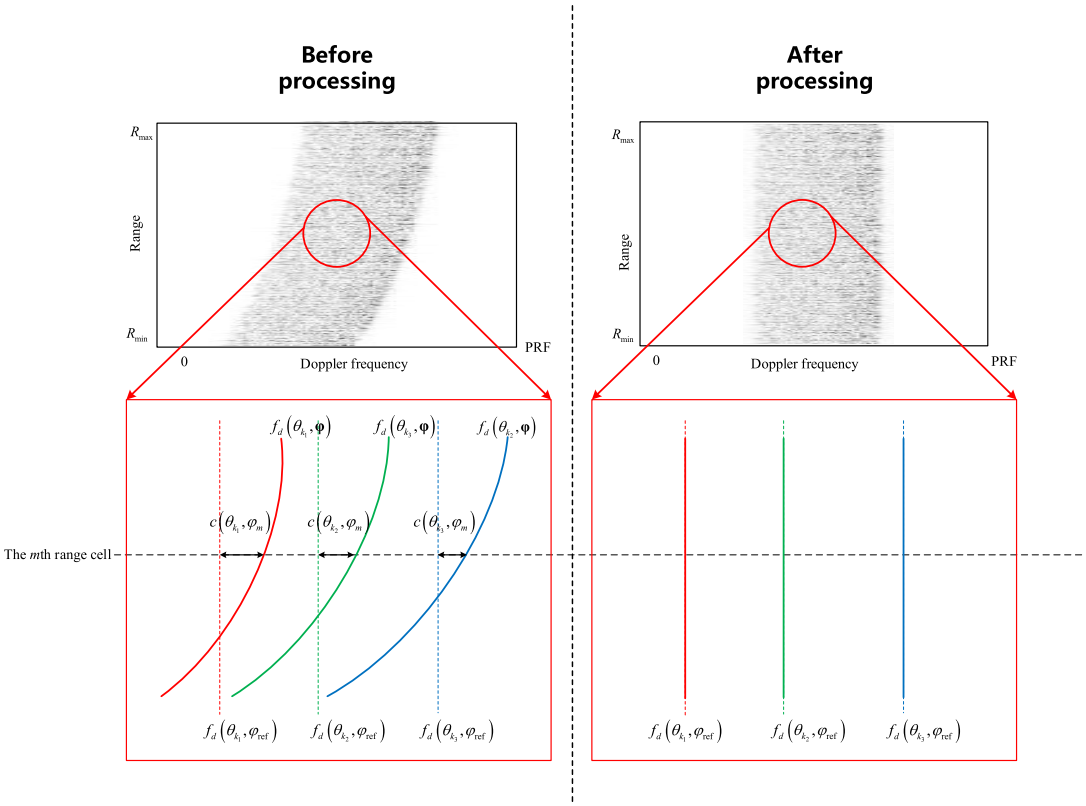


Fig. 4. Illustration of Doppler centroid compensation.

frequency, $f_d(\theta_k, \varphi_{\text{ref}})$, and the Doppler frequency, $f_d(\theta_k, \varphi_m)$, in the matrix $\mathbf{Y}_{\text{RD}}$ can be obtained by

$$c(\theta_k, \varphi_m) = [f_d(\theta_k, \varphi_m) - f_d(\theta_k, \varphi_{\text{ref}})]\frac{N_{\text{FFT}}}{\text{PRF}} \tag{33}$$

for each θ_k and φ_m with $k = 1, \ldots, K$ and $m = 1, \ldots, M$.

Once the offset is determined, it is reasonable to correct the Doppler frequency, $f_d(\theta_k, \varphi_m)$, to the reference frequency, $f_d(\theta_k, \varphi_{\text{ref}})$, by compensating the offset. Let $\mathbf{Y}'_{\text{RD}}$ denote a new matrix with the same size as $\mathbf{Y}_{\text{RD}}$. In accordance with the obtained offset value $c(\theta_k, \varphi_m)$, the entries of $\mathbf{Y}_{\text{RD}}$ can be mapped into $\mathbf{Y}'_{\text{RD}}$ using a point-by-point mapping formed as

$$\mathbf{Y}'_{\text{RD}}[m, \xi(\theta_k) - c(\theta_k, \varphi_m)] = \mathbf{Y}_{\text{RD}}[m, \xi(\theta_k)] \tag{34}$$

where

$$\xi(\theta_k) = \left\lfloor \frac{f_d(\theta_k, \varphi_m)}{\text{PRF}} N_{\text{FFT}} \right\rceil \tag{35}$$

with $\lfloor \cdot \rceil$ denoting the rounding operation toward nearest integer. In this manner, one may obtain a space-invariant spectrum by mapping all the entries of $\mathbf{Y}_{\text{RD}}$ to $\mathbf{Y}'_{\text{RD}}$, as illustrated in Fig. 4.

Because the offset value defined by (35) varies with a different angle θ_k, the mapping process shown by (34) would be formed irregularly. As a result, part of the entries in $\mathbf{Y}'_{\text{RD}}$ may be undefined, and the complex values in the Doppler spectrum become discontinuous. To overcome this problem, a 2-D interpolation procedure, such as a nearest-neighbor interpolation, a bilinear interpolation, or a cubic interpolation can be introduced to reconstruct the undefined entries in $\mathbf{Y}'_{\text{RD}}$.

C. Fast 2-D Inverse-Scattering Reconstruction

With the Doppler centroid compensation, one obtains a new matrix $\mathbf{Y}'_{\text{RD}}$ with space-invariant Doppler spectrum. By forming the inverse FFT (IFFT) of $\mathbf{Y}'_{\text{RD}}$ in the cross-range direction, the compensated echo matrix $\mathbf{Y}'$ is formed. Let

$$\mathbf{Y}' = \begin{bmatrix} y'(1,1) & y'(1,2) & \cdots & y'(1,N) \\ y'(2,1) & y'(2,2) & \cdots & y'(2,N) \\ \vdots & \vdots & \ddots & \vdots \\ y'(M,1) & y'(M,2) & \cdots & y'(M,N) \end{bmatrix} = \begin{bmatrix} \mathbf{y}'^T_1 \\ \mathbf{y}'^T_2 \\ \vdots \\ \mathbf{y}'^T_M \end{bmatrix}. \tag{36}$$

Then, the echo in the cross-range direction may be modeled as

$$\mathbf{y}'_m = \mathbf{A}_{\text{ref}}\mathbf{s}_m + \mathbf{e}'_m \tag{37}$$

for $m = 1, \ldots, M$, where

$$\mathbf{A}_{\text{ref}} = \mathbf{H} \odot \mathbf{D}_{\text{ref}} \tag{38}$$

and

$$\begin{aligned} \mathbf{D}_{\text{ref}} &= [\mathbf{d}_{\text{ref}}(\theta_1), \mathbf{d}_{\text{ref}}(\theta_2), \ldots, \mathbf{d}_{\text{ref}}(\theta_K)] \\ \mathbf{d}_{\text{ref}}(\theta_k) &= \exp\{-j2\pi f_d(\theta_k, \varphi_{\text{ref}})\mathbf{t}\}. \end{aligned} \tag{39}$$

By comparing (13) and (37), it is clear that the reflectivity vector, $\mathbf{s}_m$, can be equivalently reconstructed using the new echo matrix, $\mathbf{Y}'$, and the space-invariant matrix, $\mathbf{A}_{\text{ref}}$. For example, one may reconstruct $\mathbf{s}_m$ using l_2-norm-based regularization by replacing the space-variant terms in (23) with $\mathbf{y}'_m$ and $\mathbf{A}_{\text{ref}}$, such that

$$\hat{\mathbf{s}}_m = \left(\lambda\mathbf{I} + \mathbf{A}^H_{\text{ref}}\mathbf{A}_{\text{ref}}\right)^{-1}\mathbf{A}^H_{\text{ref}}\mathbf{y}'_m. \tag{40}$$

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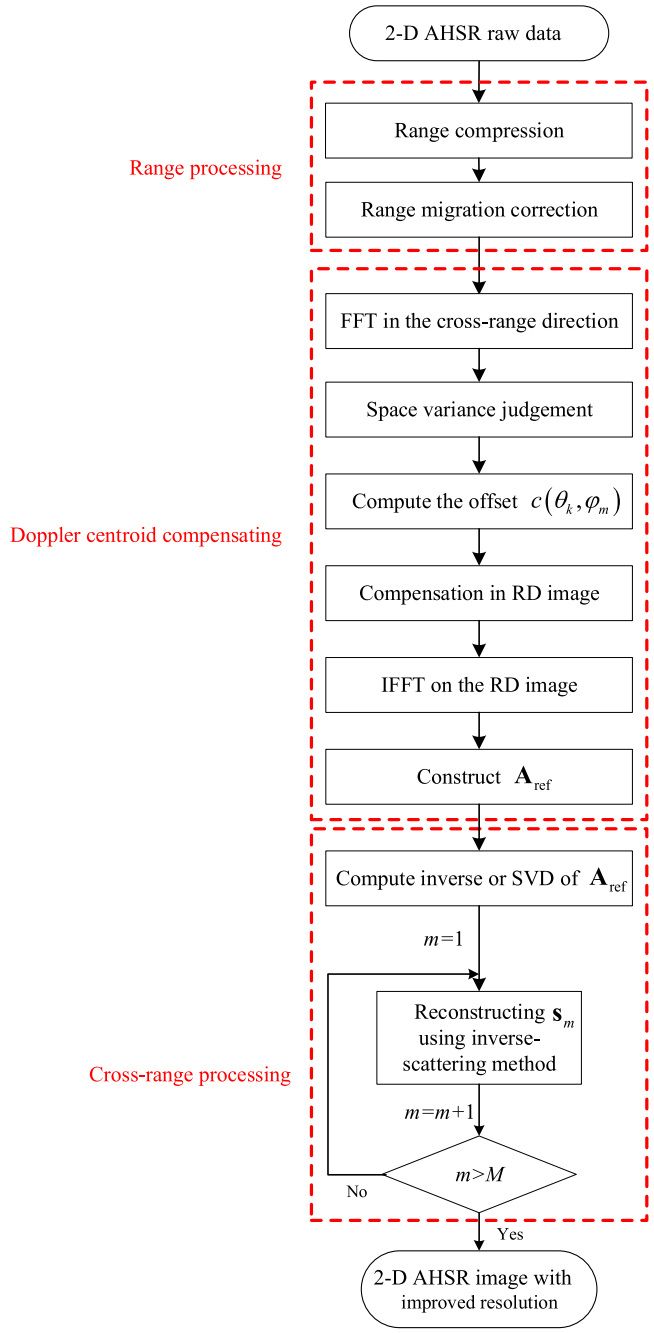


Fig. 5. Flowchart of proposed fast strategy.

It should be stressed that with the updated forward matrix $\mathbf{A}_{\text{ref}}$, the computationally extensive operations, such as any matrix inversion, will only be required to be formed once for all the echo data, $\mathbf{y}'_m$, for $m = 1, \ldots, M$, independent of the number of considered range cells, yielding a solution with significantly higher computational efficiency as compared to the brute-force inverse-scattering method in (23). A similar implementation can also be formed for other inverse-scattering methods, such as the method based on TSVD, l_1-norm regularization, and other state-of-the-art methods [18], [19], [38], [39], [41], which highly depend on the high amount of matrix operations related to the space-variant forward matrix. The proposed algorithm flowchart is illustrated in Fig. 5.

D. Complexity Analysis

Table I shows the algorithm complexity comparison between the discussed conventional brute-force (range-by-range) processing methods and the proposed fast methods. Let P denote the computational complexity of the interpolation operation involved in the proposed fast method. Notice that the operational complexity P depends on the adopted interpolation method. For example, the computational complexity of the simplest nearest-neighbor interpolation is negligible and can be omitted in the comparison, whereas Newton interpolation and other precise interpolation methods may require more computational time.

From Table I, it can be seen that the conventional inverse-scattering methods need to generate M different forward matrices, $\mathbf{A}_m$, for all range cells. As a result, the computationally extensive operations need to be performed M times. In contrast, for the proposed fast method, we only need to generate a unified forward matrix $\mathbf{A}_{\text{ref}}$ and then reconstruct the surface map of reflectivity for all range cells. This makes the operational efficiency of these methods decrease from $\mathcal{O}(MN^3)$ to $\mathcal{O}(N^3)$.

Although the computational complexity has been reduced significantly with Doppler centroid compensation, the matrix inversion is still required. Fortunately, one may instead use various powerful tools for the iterative solution of linear systems of equations to obtain the reconstruction result, such as the well-behaved steepest descent algorithms. As the related research has been well developed, herein we will not discuss these aspects in further detail.

V. Algorithmic Consideration

In this section, we proceed to examine some implementation aspects of the presented fast inverse-scattering strategy, discussing the problem of Doppler parameter estimation, as well as online processing for 2-D inverse-scattering reconstruction.

A. Doppler Parameter Estimation

As seen from (39), the Doppler centroid parameters, such as the velocity and the pitch angle, are required to construct the forward matrix, $\mathbf{A}_{\text{ref}}$. Practically, the Doppler centroid parameters can be delivered by navigational devices. However, the measuring precision of the navigational device is usually limited, which cannot satisfy the requirement for accurate Doppler centroid estimation. There are some other methods that can be incorporated to enhance the accuracy of the Doppler parameters.

A variety of data-dependent methods can be incorporated with the proposed fast inverse-scattering strategy to obtain high-quality AHSR images with low-accuracy Doppler centroid parameters. In [50] and [4], a correlation Doppler estimator (CDE) that takes advantage of the centroid symmetry of the azimuth energy spectrum is proposed. The CDE method can meet the Doppler parameter estimation tasks of imaging methods under side-view and high-squint-view conditions. In [51] and [52], a Doppler centroid estimation method based on Radon transform using the relationship between migration

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TABLE I
COMPARISON OF COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS

Method	Computational complexity of conventional methods	Computational complexity of proposed methods
TSVD	$MN^3 + (k+3)MN^2 + 2MN$	$N^3 + (M+k+2)N^2 + (k^2+2)N + P$
l_2-based method	$3MN^3 + 3MN^2 + 2MN$	$3N^3 + (M+2)N^2 + 2N + P$
l_1-based method	$3MN^3 + (3q+4)MN^2 + (q+2)MN$	$3N^3 + (5+qM+M)N^2 + (qM+4M+1)N + P$

and centroid frequency is proposed. This method is suitable for scenes containing strong scatterers. A Doppler centroid estimation method based on energy balancing is proposed in [53], [54]. This method is more suitable for uniform scenes or point targets in the high-squinting area. In [3], a minimum-entropy Doppler estimator (MEDE) method using the relation between DBS imaging entropy and Doppler centroid frequency was proposed. This method can enhance the adaptability of the scene, but it is not applicable to large high-squint areas with a poor imaging resolution. A Doppler centroid estimation method based on edge detection and fitting (EDF) was proposed in [55]. This method first detects the edge of the looking scatterer in the Doppler domain and then uses the minimum mean square error (MMSE) fitting algorithm to smooth the edges to accurately estimate the Doppler centroid.

Generally, there are many Doppler parameter estimation methods that can be used to solve the problem of large errors resulting from poor equipment parameter estimation. In practice, suitable methods can be selected according to the characteristics of the scenario.

B. Online Processing

The Doppler bandwidth is related to the field of view, such that

$$B = \frac{2v\cos\varphi}{\lambda}|\sin\theta_1 - \sin\theta_K|. \tag{41}$$

Therefore, in the case of wide-area scanning, the Doppler bandwidth is usually large. In batch processing, a large PRF is required to avoid RD spectrum overlapping, which increases the hardware burden and decreases the imaging distance. Moreover, batch processing suffers from considerable computational cost. Batch processing needs to process the data with high dimension, and the processor must wait to collect the measurement completely, which also requires additional time.

In practice, the radar transmits the signal periodically, and the received echo data is obtained in a stream of measurements. Therefore, it is desirable to derive an online implementation of the fast inverse-scattering method.

To address the problems mentioned above, we present a scheme of online processing of the proposed fast inverse-scattering method. The entire procedure is shown in Fig. 7. By means of a sliding window, the echo is truncated in the cross-range direction, and only this narrow bandwidth data is processed at a given time. The truncated echo

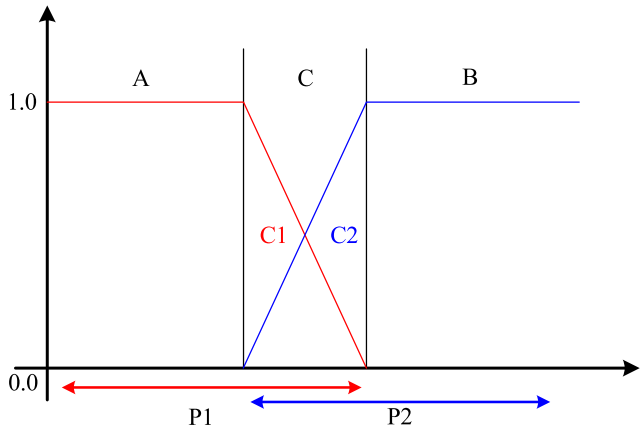


Fig. 6. Image stitching diagram.

can be expressed as

$$\mathbf{Y}_W(n) = \begin{bmatrix} y(1,n) & y(1,n+1) & \cdots & y(1,n+W-1) \\ y(2,n) & y(2,n+1) & \cdots & y(2,n+W-1) \\ \vdots & \vdots & \ddots & \vdots \\ y(M,n) & y(M,n+1) & \cdots & y(M,n+W-1) \end{bmatrix} \tag{42}$$

where n is the azimuth variable sampled by step size, and W denotes the sliding window length. In this way, the Doppler bandwidth of the truncated echo can be decreased to (10), which relaxes the requirement on high PRF.

In general, the window length of the data boundary can be selected as $1.5L$ or $2L$, with the step size being set to $0.5L$ or L. In this way, the segmented echoes can be processed by the proposed method to obtain real-time results. Finally, the imaging results of these blocks can be stitched together one by one to obtain the entire imaging result.

In this work, a stitching approach based on a linear transition of the overlapping regions is employed to achieve real-time image stitching. Assuming that $\mathbf{S}_W(n)$ and $\mathbf{S}_W(n+1)$ are adjacent results to be stitched, containing the overlapping area, $\mathbf{C}$, as shown in Fig. 6.

The stitched image can then be expressed as

$$I(x,y) = \begin{cases} A(x,y), (x,y) \in A \& (x,y) \notin B \\ w_1 * C1(x,y) + w_2 * C2(x,y), (x,y) \in A \cap B \\ B(x,y), (x,y) \in B \& (x,y) \notin A \end{cases} \tag{43}$$

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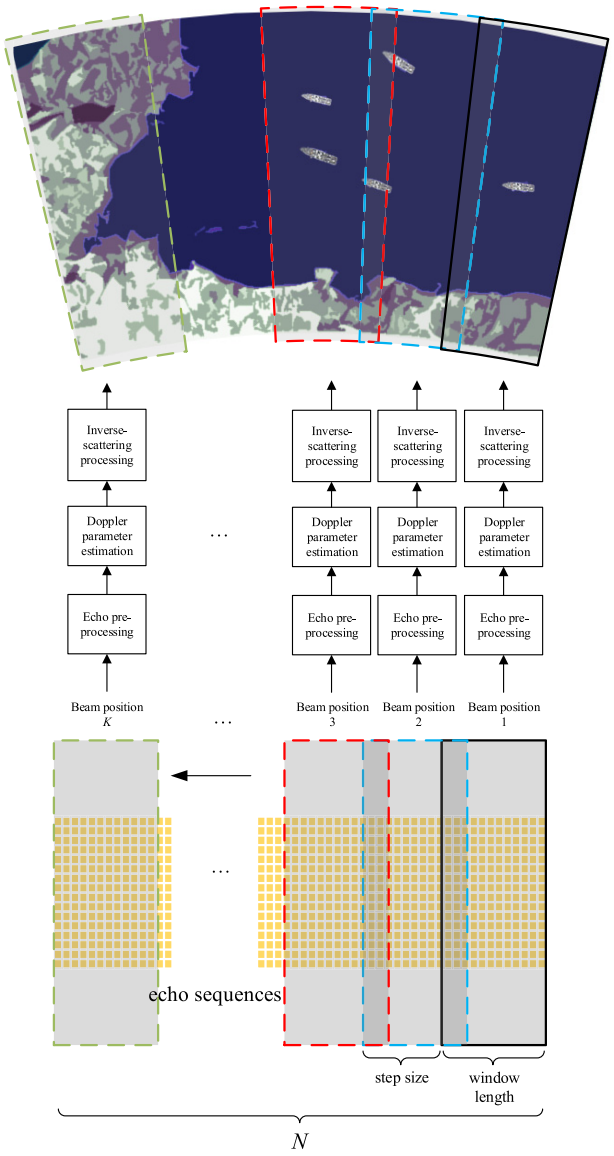


Fig. 7. Illustration of the used online processing.

where w_1 and w_2 are weights satisfying $w_1 + w_2 = 1$ with both w_1 and w_2 being bounded between 0 and 1. In practical applications, the form of the weights may instead be selected reasonably according to the characteristics of the data.

VI. Simulation Results

In this section, the simulated results are given to verify the advantage of the proposed fast strategy. First, we compare the reconstruction results produced by several alternative methods, including the TSVD method, the l_2- and the l_1-based methods. Next, we further verify the resolution performance quantitatively. Finally, we compare the computational times required by the brute-force and proposed fast methods.

Here, we consider the signal corrupted by an AWGN, with the signal-to-noise ratio (SNR) defined as

$$\mathrm{SNR} = 10\log_{10}\frac{P_s}{\sigma^2} \tag{44}$$

TABLE II
Simulation Parameters

Parameter	Value
Carrier frequency	10.5 GHz
Signal bandwidth	3.6 MHz
Pulse width	2 μs
Scanning range	5 °-20 °
Scanning speed	90 °/s
Beam width	4 °
PRF	4 kHz
Platform height	1000 m
Airborne platform speed	120 m/s

where P_s represents the peak power of the raw echo and σ^2 is the variance of the noise. The used simulation parameters are shown in Table II. Our simulation platform are conducted using a 64-bit MATLAB utilizing only one CPU core of a workstation with i7 10700 3.4 GHz CPU and 16 GB RAM.

A. Inverse-Scattering Results

A typical image including several ships is used as a simulation scenario, as shown in Fig. 8(a). Moreover, some point targets are set at the top of Fig. 8(a) for the quantitative analysis of resolution. For each true target, we independently add a random phase that follows uniform distribution on the interval $(0, 2\pi]$. Fig. 8(b) shows the real beam image with SNR = 20 dB after pulse compression in range, which suffers from low cross-range resolution. Fig. 8(c) illustrates the result of the DBS method. We use a rectangular window function as the weighting function in the FFT. It is clear that the DBS method can enhance the cross-range resolution, but the improvement is still limited. The TSVD method can obtain higher cross-range resolution than the conventional DBS method, and the outline of the surface target is sharpened, as shown in Fig. 8(d). However, the TSVD method suffers from severe transverse ripples. In contrast, the l_2-based method improves the image quality, as shown in Fig. 8(e). The transverse ripples are well suppressed, outperforming the TSVD method. Both the TSVD and l_2 method overcome the deficiencies of the traditional DBS method with limited resolution. Fig. 8(f) shows the results of the l_1-based method. Due to the sparsity of the scenario, the result of the l_1-based method is obviously better than the former methods. The ship targets contour is well recovered. Compared with the traditional DBS method, the resolution of the l_1-based method is significantly improved.

Notice that the results shown in Fig. 9(d)–(f) have been obtained using the brute force implementation, processing the echo matrix range-by-range. These calculations require 13.99, 17.86, and 18.68 s to formulate the reconstruction results of size 190×667 (range $\times$ cross-range), respectively. In contrast, the proposed fast strategy based on Doppler centroid compensation only requires 0.084, 0.117, and 0.155 s to formulate the results with marginal resolution loss, as shown in Fig. 8(g)–(i). The resolution improvement of the inverse-scattering methods can be further illustrated by the profiles of the azimuth cut located at the 172nd range cell shown in Fig. 9.

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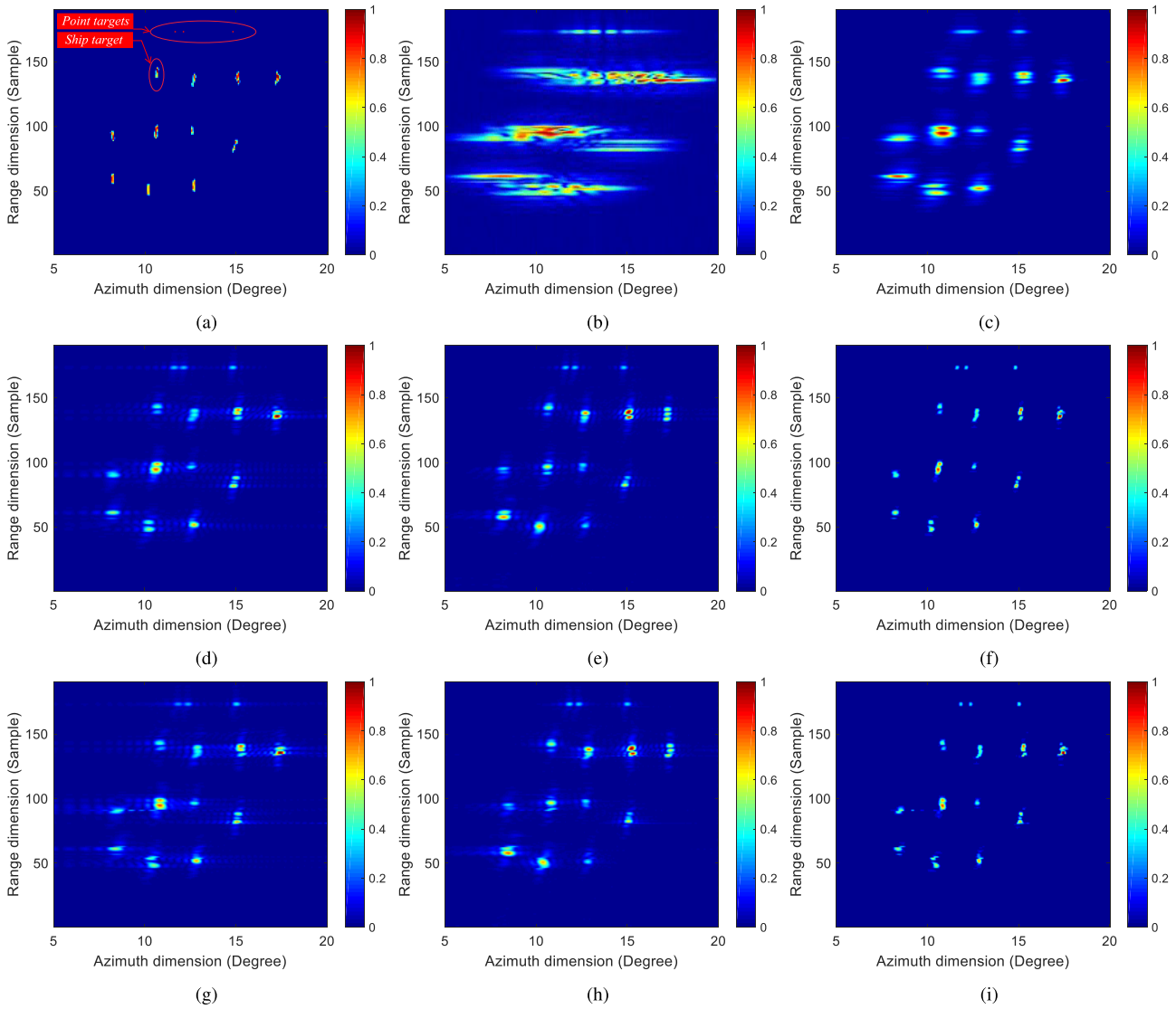


Fig. 8. Simulated results for a 2-D scenario. (a) Truth. (b) Real beam echo after matched filtering. (c) Conventional DBS method. (d) Brute force TSVD method. (e) Brute force l_2-based method. (f) Brute force l_1-based method. (g) Fast TSVD method. (h) Fast l_2-based method. (i) Fast l_1-based method.

B. Resolution Comparison

To quantitatively analyze the resolution performance, both the 3 dB width and peak-to-sidelobe ratio (PSLR) are introduced. The PSLR is defined as

$$\mathrm{PSLR} = 10\log_{10}\left(\frac{P_{sl}}{P_{pk}}\right) \tag{45}$$

where P_{pk} and P_{sl} denote the peak power of the mainlobe and the highest sidelobe in the vicinity of the peak, respectively.

Table III quantitatively verifies the resolution performance of the proposed methods. It is clear that all inverse-scattering methods can obtain a higher resolution than the traditional DBS method. Meanwhile, the resolution performance of the proposed fast inverse-scattering methods is similar to that of the traditional inverse-scattering methods. It is worth mentioning that the l_1-based method can achieve the best resolution performance.

C. Computational Complexity Comparison

Fig. 10 shows the curves of computational complexity for the brute-force and proposed fast methods with constant M (recall that M represents the number of sampling points in the range dimension) and a varying N (recall that N denotes the number of sampling points in the cross-range dimension). The computational complexity is defined as the total number of operations for multiplication and division.

Assume that $M = 700$ and that N varies from 1 to 5000. It is clear that, for each kind of inverse-scattering method, the proposed fast strategy enjoys higher efficiency than its brute-force implementation. Typically, when $N = 1000$, the fast TSVD method, the l_2- and l_1-based methods reduce the required computational complexity by 420 times, 610 times, and 40 times, compared with the brute-force methods, respectively.

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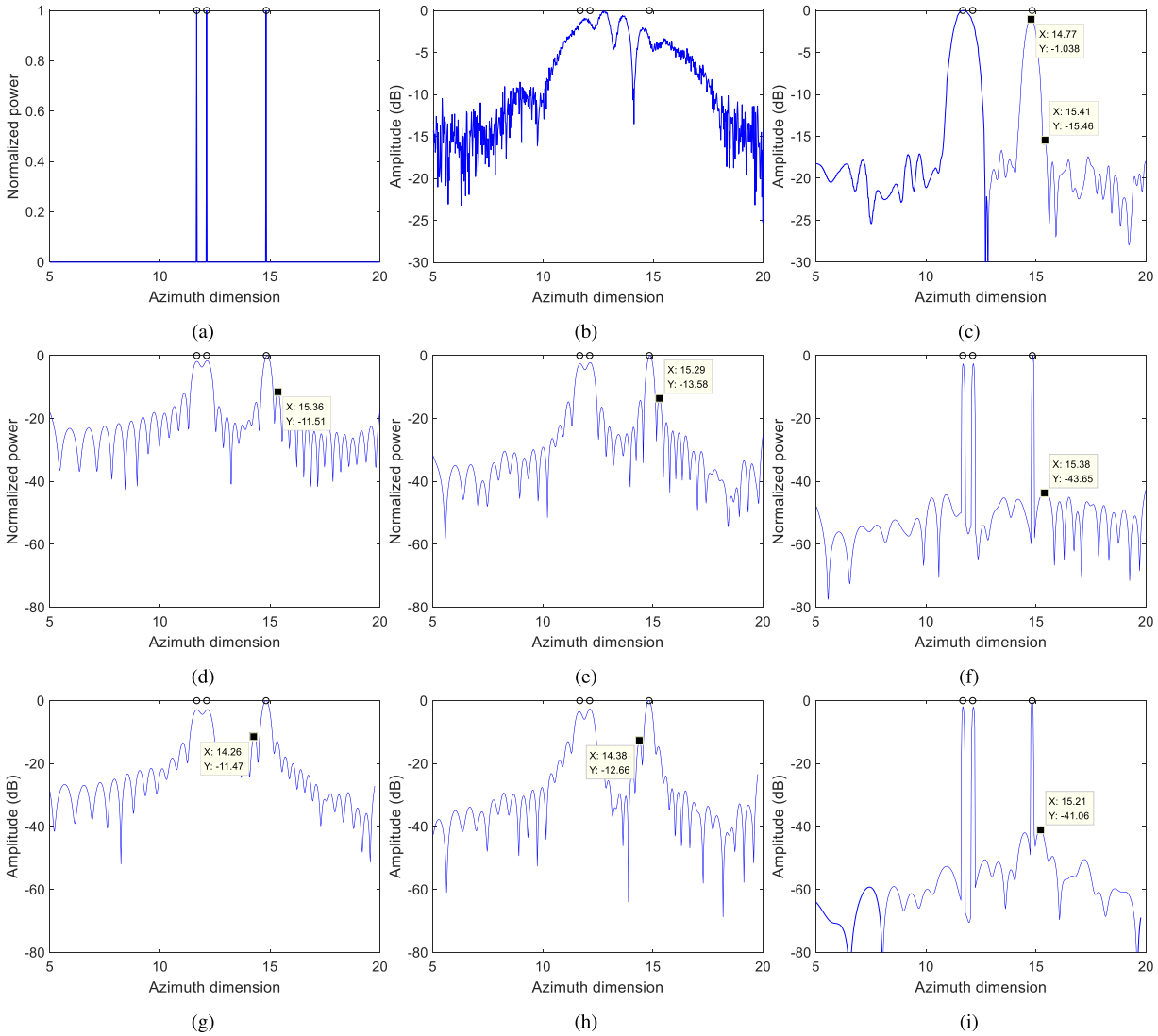


Fig. 9. 172th range cell profiles (point targets) of Fig. 8. (a) Truth. (b) Real beam echo. (c) Conventional DBS method. (d) Brute force TSVD method. (e) Brute force l_2-based method. (f) Brute force l_1-based method. (g) Fast TSVD method. (h) Fast l_2-based method. (i) Fast l_1-based method.

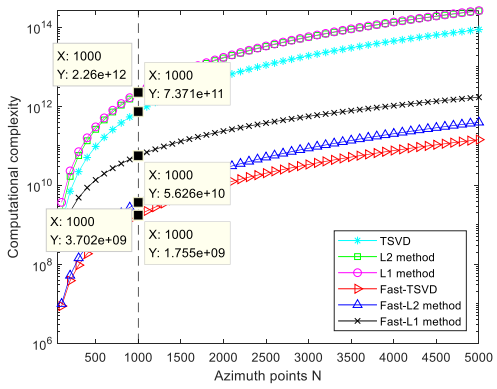


Fig. 10. Computational complexity for different methods (constant M).

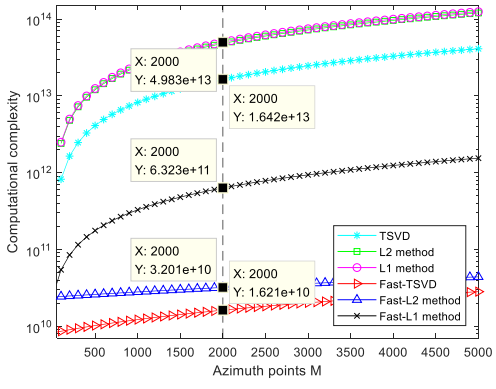


Fig. 11. Computational complexity for different methods (constant N).

Fig. 11 illustrates the curves of computational complexity for the brute-force and proposed fast methods with a constant N and a varying M. Assume that $N = 2000$ and that M varies from 1 to 5000. As before, for each inverse-scattering method, the computational efficiency of the proposed fast method is higher than the traditional brute-force implementation. Typically, when $M = 2000$, the fast TSVD method, l_2-based and l_1-based methods reduce the required computational

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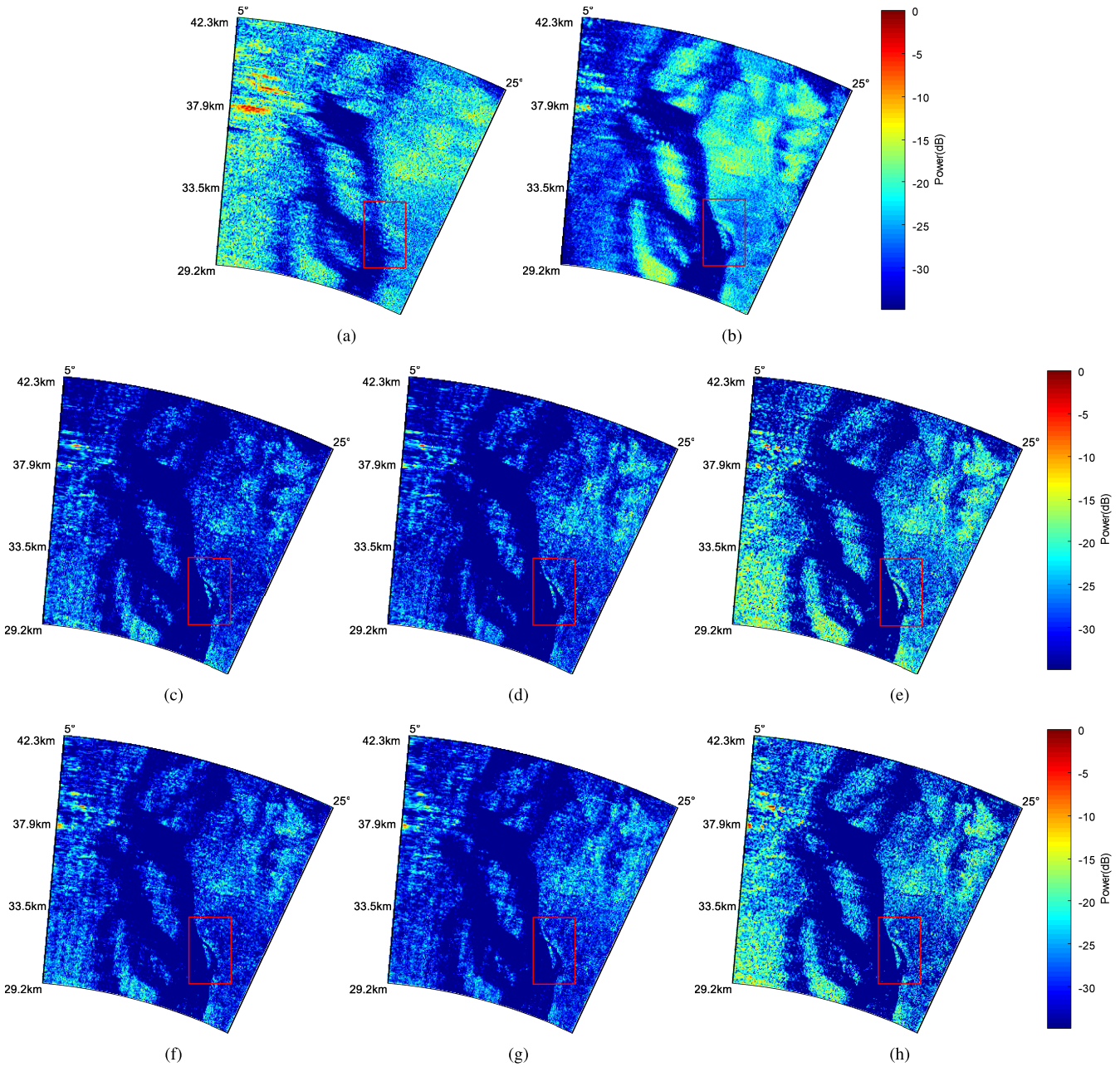


Fig. 12. Experimental results. (a) Real beam data after matched filtering. (b) Conventional DBS method. (c) Brute force TSVD method. (d) Brute force l_2-based method. (e) Brute force l_1-based method. (f) Fast TSVD method. (g) Fast l_2-based method. (h) Fast l_1-based method.

complexity by 1010 times, 1557 times, and 79 times, compared with the brute-force methods, respectively.

In conclusion, the simulated results verify that the proposed fast inverse-scattering methods significantly reduce the required computational complexity without sacrificing the resulting estimates' resolution.

VII. Real-Data Results

We proceeded to verify the advantage of the proposed method using real data. The experimental parameters are listed in Table IV.

In the experiment, we opted for a typical scenario containing a river bank environment. In the following figures, the results are plotted in dB. Fig. 12(a) shows the real beam echo after matched filtering in the range dimension. It is seen that the real beam image suffers from coarse cross-range resolution. The traditional DBS method can obtain finer cross-range resolution to some extent, as shown in Fig. 12(b). However, the resolution improvement is limited. As a result, it is still difficult to distinguish the island from the land, as shown by the area marked with red. Fig. 12(c)–(e) illustrates the results of the brute-force TSVD, l_2- and l_1-based inverse-scattering approaches, respectively. It is clear that these approaches can achieve higher resolution than the traditional DBS method. The scenario has been better reconstructed, and the outline of the river bank is noticeably sharper. However, these brute-force

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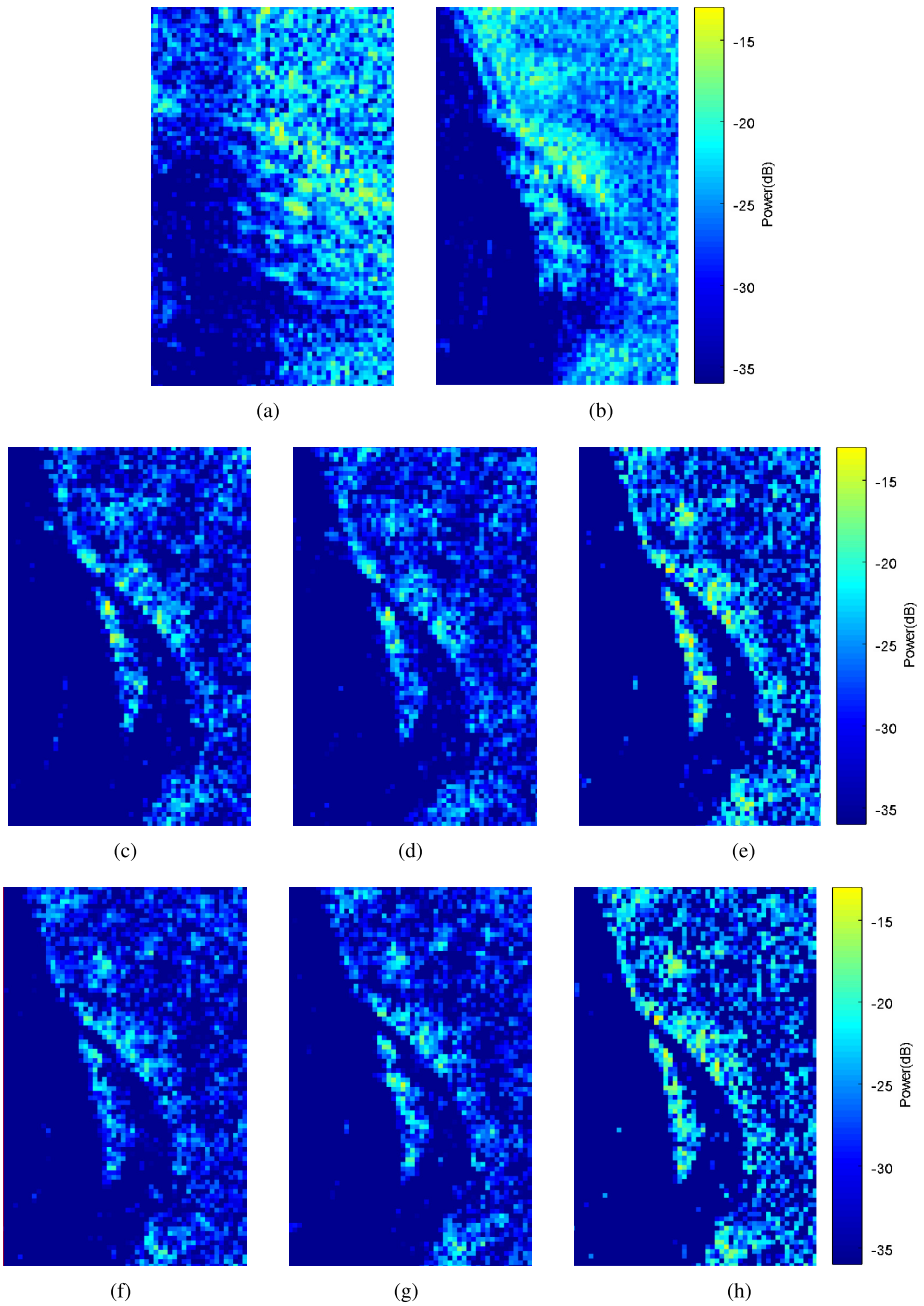


Fig. 13. Partially enlarged graphs for Fig. 12. (a) Real beam data. (b) Conventional DBS method. (c) TSVD method. (d) l_2-based method. (e) l_1-based method. (f) Fast TSVD method. (g) Fast l_2-based method. (h) Fast l_1-based method.

methods suffer from heavy computational complexity. They require 357.71, 134.61, and 243.24 s to formulate the reconstruction results of size 350×1050 (range $\times$ cross-range), respectively.

The results of the proposed fast TSVD approach and l_2- and l_1-based approaches are shown in Fig. 12(f)–(h), respectively. It can be seen that the proposed fast approaches offer the same excellent cross-range resolution as the brute-force approaches, while the computational complexity is much lower, requiring only 5.77, 4.71, and 7.88 s to formulate the reconstruction results, respectively.

In addition, the partially enlarged graphs for the riverway are shown in Fig. 13, which further demonstrates the resolution performance of the above-mentioned inverse-scattering methods.

Note that there is no original scenario for the real data. We cannot use the indexes introduced in the simulation to evaluate the image quality quantitatively. Instead, we may use image entropy to evaluate the resolution of a radar image [3], due to the fact that this entropy well describes the degree of sharpness of data or images. Herein, it is employed to quantitatively evaluate the quality of the real-data results. The image entropy of Fig. 13 is listed in Table V. It is seen that the image qualities of Fig. 13(c)–(h) are better than that of Fig. 13(b), with lower entropy. This means that the resolution of the 2-D inverse-scattering reconstruction algorithm is higher than that

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TABLE III
COMPARISON OF RESOLUTION PERFORMANCE

Method	3 dB width	PSLR
DBS	0.64°	-14.42 dB
TSVD	0.31°	-11.51 dB
l_2-based	0.29°	-13.58 dB
l_1-based	0.13°	-43.65 dB
Fast TSVD	0.33°	-11.47 dB
F-l_2-based	0.30°	-12.66 dB
F-l_1-based	0.13°	-41.06 dB

TABLE IV
PARAMETER CONDITIONS OF THE MEASURED DATA

Parameter	Value
Carrier frequency	X-band
Bandwidth	3.6 MHz
Pulse width	32 μs
Scanning range	5 °-25 °
Scanning speed	5 °/s
Beam width	2.2 °
Pulse repetition interval (PRI)	512 μs
Platform height	2620 m
Airborne platform speed	143 m/s
Data size (range×cross-range)	350×1050

TABLE V
COMPARISON OF IMAGE QUALITY

Method	Entropy
Real data	5.71
Conventional DBS	5.45
TSVD	5.12
l_2-based	4.83
l_1-based	4.28
Fast TSVD	5.16
Fast l_2-based	4.89
Fast l_1-based	4.24

of the conventional FFT-based DBS method. Moreover, it is worth noting that the l_1-based method again yields the highest image quality among three inverse-scattering methods.

It should also be noted that the entropy values obtained by traditional brute-force and proposed fast inverse-scattering methods are quite close to each other. It can be concluded that the proposed method is as good as the traditional brute-force method in terms of imaging quality.

In conclusion, the measured data results verify that the proposed fast inverse-scattering strategy greatly improves operational efficiency with marginal performance loss.

VIII. CONCLUSION

In this article, we propose a fast 2-D inverse-scattering reconstruction strategy for AHSR imaging. The resulting method greatly decreases the computational complexity required by the inverse-scattering reconstruction method. The simulated and real data results indicate that the proposed fast strategy provides higher computational efficiency while maintaining notable resolution improvement.

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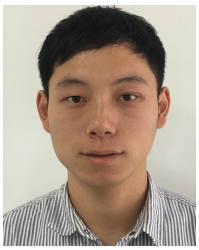
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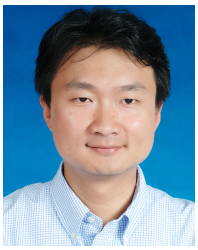
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